

# Foundations of Robotics

## Homework 3.5 (optional)

1. Convert the ZYX Euler angles  $(\alpha, \beta, \gamma) = (\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3})$  to a rotation matrix. (Note:  $\alpha, \beta, \gamma$  are in the order of ZYX.)
2. Are Euler angles unique? Explain with an example.
3. Explain the concept of gimbal lock in the context of Euler angles, and provide an example rotation (in Euler angles) that leads to gimbal lock.
4. Write down the general form of a quaternion, and explain the conditions that a quaternion must satisfy to represent a valid rotation.
5. Convert the unit quaternion  $q = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i + 0j + 0k$  to a rotation matrix, and verify if this rotation matrix is orthonormal.
6. Two rotations are represented by unit quaternions:  $q_1 = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i + 0j + 0k$  and  $q_2 = 1 + 0i + 0j + 0k$ . Compute the spherical linear interpolation (SLERP) for  $t = 0.5$ , and verify if the resulting quaternion is normalized.